**TableS1 Quality assessment of included studies according to the Newcastle-Ottawa scales**

| Study | Representativeness of exposed cohort | Selection of nonexposed cohort | Ascertainment of exposure | Outcome of interest was not present at start of study? | Comparability of cohort on the basis of the design | Assessment of outcome | Was follow-up long enough for outcome to occur? | Adequacy of follow-up of cohorts | Quality score |
|---|---|---|---|---|---|---|---|---|---|
| Vicente et al[21] | * | * | * | * | * | * | * | - | 7 |
| Takeuchi et al[22] | * | * | * | * | - | * | * | - | 6 |
| Fujiya et al[7] | * | * | * | * | * | * | * | * | 8 |
| Powell et al[5] | * | * | * | * | - | * | * | * | 7 |
| Tu et al[11,25] | * | * | * | * | - | * | * | - | 6 |
| Li et al[23] | * | * | * | * | * | * | * | * | 8 |
| Kader et al[18] | * | * | * | * | * | * | * | - | 7 |
| Hayashi et al[6] | * | * | * | * | - | * | * | * | 7 |
| Kubota et al[4] | * | * | * | * | * | * | * | - | 7 |
| Li et al[16] | * | * | * | * | - | * | * | - | 6 |
| Watanbe et al[26] | * | * | * | * | * | * | * | - | 7 |
| Saito et al[13] | * | * | * | * | - | * | * | - | 6 |
| Climent et al[19] | * | * | * | * | * | * | * | * | 8 |
| Tokunaga et al[17] | * | * | * | * | * | * | * | - | 7 |
| Mohri et al[24] | * | * | - | * | - | * | * | * | 6 |
| Jiang et al[20] | * | * | - | * | * | * | * | - | 6 |
| Jin et al[29] | * | * | * | * | * | * | * | - | 7 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Suzuki et al[35] | * | * | * | * | * | * | * | - | 7 |
| Nagasako et al[32] | * | * | * | * | - | * | * | - | 6 |
| Yoo et al[9] | * | * | * | * | * | * | * | - | 7 |
| Sierzega et al[8] | * | * | * | * | - | * | * | - | 6 |
| Andreou et al[31] | * | * | * | * | * | * | * | - | 7 |
| Kiuchi et al[10] | * | * | * | * | - | * | * | - | 6 |
| Kim et al[33] | * | * | * | * | - | * | * | - | 6 |
| Liu et al[34] | * | * | * | * | - | * | - | * | 6 |
| Eto et al[27] | * | * | * | * | * | * | * | - | 7 |
| Tsujimoto et al[30] | * | * | * | * | - | * | * | * | 7 |
| Zhuang et al[28] | * | * | * | * | - | * | * | - | 6 |
| Kamarajah et al[36] | * | * | * | * | * | * | * | - | 7 |
| Kanda et al[37] | * | * | * | * | - | - | * | * | 6 |
| Tokunaga et al[38] | * | * | * | * | * | * | - | - | 6 |
| Wu et al[12] | * | * | - | * | * | * | * | - | 6 |

Each asterisk represents if individual criterion within the subsection were fulfilled.

## Supplemental material Meta-regression analysis for OS

Meta-regression — Number of obs = 18
REML estimate of between-study variance — tau2 = .03339
% residual variation due to heterogeneity — I-squared_res = 46.78%
Proportion of between-study variance explained — Adj R-squared = 6.53%
Joint test for all covariates — Model F(7,10) = 1.20
With Knapp-Hartung modification — Prob > F = 0.3834

| lnhr | Coef. | Std. Err. | t | P>\|t\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| POC_rate | -.2006577 | .1577477 | -1.27 | 0.232 | -.5521416 | .1508261 |
| Occurrence_time | -.2351851 | .1638558 | -1.44 | 0.182 | -.6002785 | .1299083 |
| Study_quality | .0013148 | .1756559 | 0.01 | 0.994 | -.3900708 | .3927005 |
| CD_grade | -.0144206 | .1891127 | -0.08 | 0.941 | -.4357899 | .4069488 |
| Stage | .2862037 | .1989511 | 1.44 | 0.181 | -.1570869 | .7294943 |
| Region | .2641502 | .2400703 | 1.10 | 0.297 | -.2707597 | .79906 |
| sample_size | -.1335684 | .1639563 | -0.81 | 0.434 | -.4988858 | .2317491 |
| _cons | .40503 | .9184436 | 0.44 | 0.669 | -1.64139 | 2.45145 |